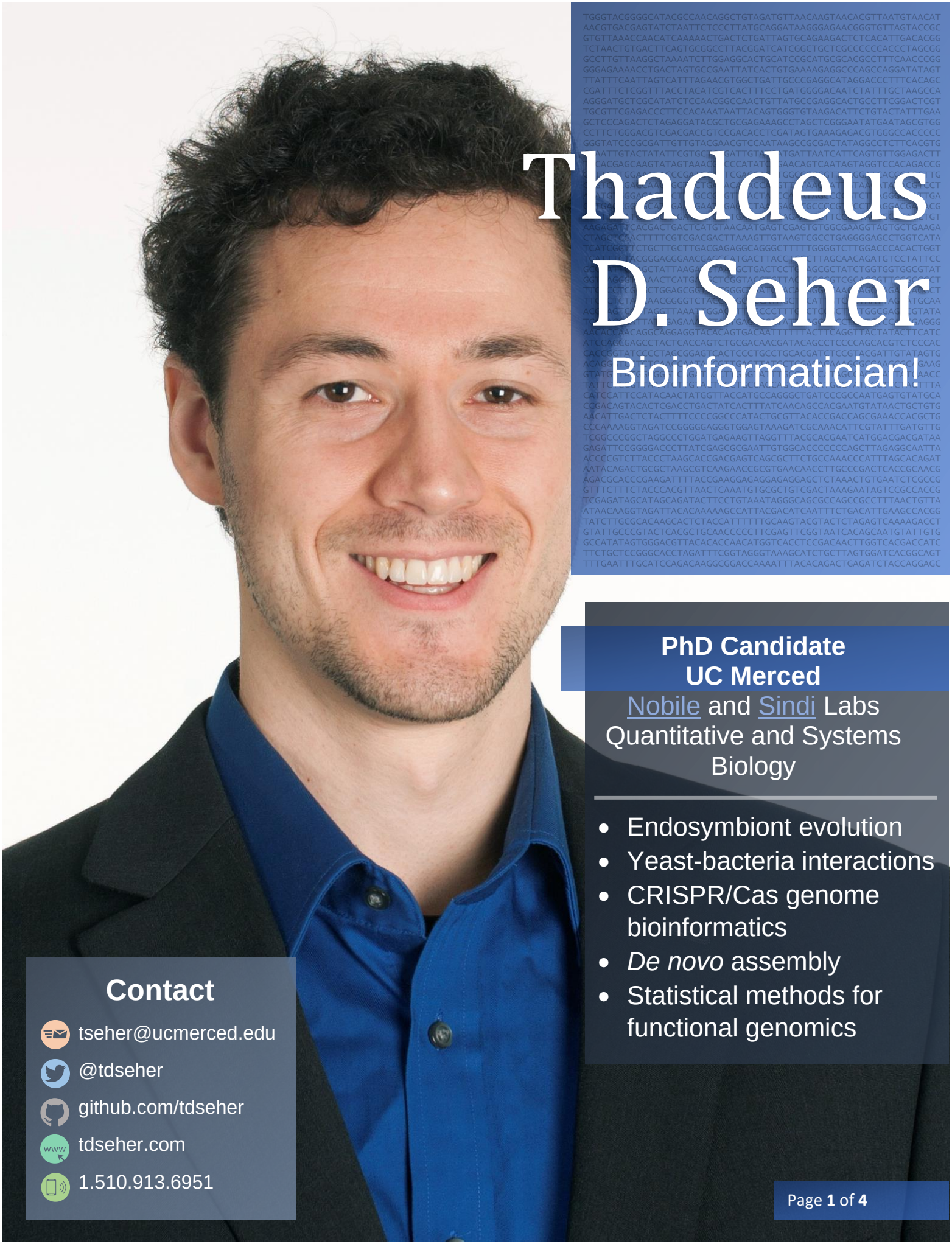

Thaddeus D. Seher

Bioinformatician!

PhD Candidate
UC Merced

Nobile and Sindi Labs
Quantitative and Systems Biology

- Endosymbiont evolution
- Yeast-bacteria interactions
- CRISPR/Cas genome bioinformatics
- *De novo* assembly
- Statistical methods for functional genomics

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Bioinformatician!

Statistics

Today's most important political, economic, social, and technological advances depend on working with data. Statistics provides a mathematical summarization of the most important parts of information, regardless of its source. I want my students to perceive the underlying numerical explanations for events, and I want my students to master statistics so they can have the tools necessary to significantly affect the world in a good way.

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Education

Aug 2015 – Present
PhD Candidate
Quantitative and Systems Biology, UC Merced

Sept 2006 – June 2010
BS: Genetics
Minor: Quantitative Biology and Bioinformatics
Biological Sciences, UC Davis

Work Experience

May 2013 – July 2015
Computer Resource Specialist
Ludington Laboratoty (MCB), UC Berkeley
- Nitrate metabolizing Planctomycetes mediate fertilizer contamination in groundwater.
- Methods for detecting horizontal gene transfer in metagenomes.

Barton Laboratory (MCB), UC Berkeley
- 16S amplicon sequencing of anti-immunoglobulin-specific commensal gut microbiome.
- Ecological associations of mouse microbial gut community through time.
- Mouse TLRs influence gut microbial communities.
- Cohousing litter mates is an effective strategy for gut microbiome control.

Oct 2010 – Oct 2012
Junior Specialist
Kopp Laboratory (Evolution), UC Davis
- Transcriptome sequencing and genetic mapping of *Drosophila bipectinata* sex comb gene.
- Massively-parallel sequencing reveals the female-specific abdominal pigmentation gene in *Drosophila serrata*.

Dec 2008 – Oct 2010
Programming Assistant
Langley Laboratory (Evolution), UC Davis
- Illumina sequencing machine administrator, quality control pipeline, and SRA submissions.

Honors + Awards

Aug 2016
Program for Excellence in Science 3-year Sponsorship
AAAS

March 2017
NSF Graduate Research Fellowship Program

March 2016
Honorable Mention
NSF

April 2016
Graduate Award
UC Merced

April 2016
Quantitative and Systems Biology Summer Fellowship
UC Merced

Jan 2016
Graduate Fellowship Incentive Program Award
UC Merced

Fall 2006 – Spring 2009
Davis Honors Challenge
UC Davis

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Bioinformatician!

Research

I am a Molecular Geneticist and Bioinformatician studying metagenomics, comparative genomics, and endosymbiosis. I specializes in library prep, assembly, and annotation. I computationally model molecular processes like DNA splicing and genomic evolution. Necessarily, my inquiries revolve around numerical algorithms, graph theory, and optimization. Along the way I wring as much data as possible from high throughput sequencing data through quantification of uncertainty.

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Publications

William Ludington, **Thaddeus Seher**, Olin Applegate, Xunde Li, Joseph Kliegman, Charles Langelier, Edward Atwill, Thomas Harter, Joseph DeRisi. Assessing biosynthetic potential of agricultural groundwater through metagenomic sequencing: A diverse anammox community dominates nitrate-rich groundwater. *PLoS ONE*: December 2016 12(4), e0174930 doi: 10.1371/journal.pone.0174930 PMID: 28384184.

Amir Yassin, Emily Delaney, Adam Reddiex, **Thaddeus Seher**, Héloïse Bastide, Nicholas Appleton, Justin Lack, Jean David, Stephen Chenoweth, John Pool, Artyom Kopp. The *pdm3* locus is a hotspot for recurrent evolution of female-limited color dimorphism in *Drosophila*. *Current Biology*: August 2016, pii S0960-9822(16)30767-9 doi: 10.1016/j.cub.2016.07.016 PMID: 27546577.

Meghan Koch, Gabrielle Reiner, Kyler Lugo, Lieselotte Kreuk, Alison Stanbery, Eduard Ansaldo, **Thaddeus Seher**, William Ludington, Gregory Barton. Maternal IgG and IgA antibodies dampen mucosal T helper cell responses in early life. *Cell*: May 2016 165(4), 827-841 doi: 10.1016/j.cell.2016.04.055 PMID: 27153495.

Sarah Signor, **Thaddeus Seher**, Artyom Kopp. Genomic resources for multiple species in the *Drosophila ananassae* species group. *Fly*: January 2013 7(1), 47-56 doi: 10.4161/fly.22353 PMID: 23639891.

Thaddeus Seher, Chen Siang Ng, Sarah Signor, Ondrej Podlaha, Olga Barmina, Artyom Kopp. Genetic basis of a violation of Dollo's law: re-evolution of rotating sex combs in *Drosophila bipectinata*. **Front cover** *Genetics*: December 2012 192(4), 1465-1475 doi: 10.1534/genetics.112.145524 PMID: 23086218.

Posters + Abstracts

Meghan Koch, Gabrielle Reiner, Kyler Lugo, **Thaddeus Seher**, William Ludington, Gregory Barton. Anti-commensal IgG3 antibodies reinforce intestinal immune homeostasis. Inflammatory Bowel Diseases—Mucosal Inflammation Program: Abstract + Poster Dec 2014 doi: 10.1097/01.MIB.0000456934.57591.13

Meghan Koch, Gabrielle Reiner, Kyler Lugo, **Thaddeus Seher**, William Ludington, Gregory Barton. TLR-induced anti-commensal antibodies regulate intestinal homeostasis. Kenneth Rainin Foundation's 2014 Innovations Symposium—Taming the Microbiome: Abstract + Poster July 2014.

Thaddeus Seher, Ted Kim, Carolyn Elya, Javier Navarro, Eoin Brodie, Michael Eisen, William Ludington. Ecological epistasis in the fly gut on spatial, temporal, and evolutionary dimensions. 2013 U.C. San Francisco/U.C. Berkeley Immunology Retreat: Poster September 2013.

Thaddeus Seher, Ondrej Podlaha, Artyom Kopp. Evolution of abdominal pigmentation in the montium subgroup of *Drosophila*. 2011 Genetics Society of America *Drosophila* Research Conference: Abstract + Poster March 2011.

Thaddeus D. Seher

Bioinformatician!

Teaching

My teaching philosophy revolves around the two things strongly motivate people: the joy of learning, and the satisfaction of demonstrating mastery. Through discussion, I hint at what is possible, and encourage the students to explore whatever they conceive. Along the way, I empower them with foundational methods to help them not only achieve their immediate goals, but also to expand their fundamental understanding of statistics.

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Invited Lectures

Thaddeus Seher. Subtleties of BLAST parameters. BIO-184: Object Oriented Programming for Biologists: April 2017.

Thaddeus Seher. NumPy data structures and Matplotlib recipes. BIO-184: Object Oriented Programming for Biologists: April 2017.

Thaddeus Seher. Practical steps to forge a career in science (for undergraduates). Society for Advancement of Chicanos/Hispanics and Native Americans in Science: April 2016.

Thaddeus Seher. χ^2 tests: contingency tables, directionality, & estimators. BIO-175: Biostatistics: April 2016.

Thaddeus Seher. t-tests: α, confidence intervals, & hypothesis directionality. BIO-175: Biostatistics: March 2016.

Teaching

Fall 2018	**Teaching Assistant** BIO-018: Statistics for Scientific Data Analysis
Fall 2017	**Statistics Tutor for Graduate Students and Postdocs** Graduate Division, UC Merced
Spring 2017	**Teaching Assistant** BIO-184: Object Oriented Programming for Biologists
Fall 2017 Fall 2016	**Teaching Assistant** MATH-015: Introduction to Scientific Data Analysis
Spring 2016	**Teaching Assistant** BIO-175: Biostatistics
Fall 2015	**Teaching Assistant** BIO-001: Contemporary Biology
Spring 2014	**Mentor** Undergraduate Research Apprentice Program (L&S) UC Berkeley
Summer 2016	**STEM Instructor** 5-week workshop for pre-K to 5th grade Harvest Park Education Center

Professional Enterprises

2018 – Present	**President** Graduate Student Association, UC Merced
2016 – 2018	**Founder & Chief Technology Officer** Radio Bio Podcast, UC Merced
2016 – Present	**Founder & President** BioThought
2015 – 2017	**Vice President** Quantitative Project, UC Merced
2012 – Present	**Founder & Committee Member** BAFC Schollars College Endowment Fund

Hobbies